Data Quality Report

1. Introduction

This report assesses the quality of the ten PostgreSQL staging tables that were profiled in the data profiling report: BKPF, BSEG, CSKS, CSKT, KNA1, LFA1, SKA1, SKAT, T001 and T002. The profiling report described what the data looks like. This report goes one step further and checks the data against a set of quality rules, to decide where it is good enough to use and where it needs correction first. The tables are SAP finance extracts and together hold 1,465,129 rows.

2. Approach and Rules

The assessment uses the results already produced by the profiling notebooks (row counts, missing values, exact duplicates, data types and numeric statistics). No new queries were run. Quality is judged on four dimensions: completeness, uniqueness, validity and consistency.

Because no formal business rules were supplied with the data, the rules below were defined for this report from the column names and normal SAP conventions. They should be confirmed with the data owners before the results are treated as final.

Completeness: a field that identifies a record or describes it in a way that is needed for reporting must not be NULL (these are called mandatory fields here; optional fields such as second name lines, tax code and line item text are not scored). Uniqueness: a table must not contain rows that are identical in every column. Validity: values must be of a suitable type and within a sensible range. Consistency: the same kind of field must have the same type and name in every table.

Each failed rule is given a severity based on the share of rows affected: Low for less than 1 percent, Medium for 1 to 5 percent and High for more than 5 percent. The completeness score of a table is the average percentage of populated values over its mandatory fields, and the uniqueness score is 100 minus the percentage of duplicate rows.

3. Summary of Results

Table 1 summarises the result for each table. Of the 45 mandatory fields checked, 20 contain NULLs. Only BKPF and T002 have fully populated mandatory fields, and only SKA1 and T002 are free of duplicate rows. The weakest table is SKAT, with a completeness score of 65.09 percent because chart_of_accounts is empty in every row, and a uniqueness score of 90.80 percent. LFA1 and T001 also have uniqueness scores below 90 percent, and no table passes every rule. The Low severity for BKPF, BSEG and T002 comes only from duplicate rows or from columns stored with a poor data type (Section 4.3), and the share affected is small.

Table 1: Quality scores by table

Table	Rows	Mandatory fields failed	Completeness %	Uniqueness %	Highest severity
BKPF	150,057	0 of 15	100.00	99.69	Low
BSEG	332,106	2 of 5	99.99	99.98	Low
CSKS	3,353	3 of 3	99.97	98.81	Medium
CSKT	26,977	1 of 1	98.95	99.99	Medium
KNA1	20,209	3 of 3	99.74	99.89	Low
LFA1	3,035	3 of 3	97.17	87.18	High
SKA1	146,692	2 of 2	97.23	100.00	Medium
SKAT	782,051	3 of 3	65.09	90.80	High
T001	605	3 of 3	97.36	88.93	High
T002	44	0 of 7	100.00	100.00	Low

4. Results by Dimension

4.1 Completeness

Table 2 lists every mandatory field that contains NULLs. The mandatory fields that were checked and found fully populated are all 15 columns of BKPF, gl_account_no, account_type and debit_credit_indicator in BSEG, and all 7 columns of T002.

Table 2: Mandatory fields with NULL values

Table	Mandatory field	NULL rows	% NULL	Severity
BSEG	amount_local_currency	3	0.00	Low
BSEG	amount_document_currency	92	0.03	Low
CSKS	valid_from_date	1	0.03	Low
CSKS	company_code	1	0.03	Low
CSKS	currency_code	1	0.03	Low
CSKT	cost_center_name	282	1.05	Medium
KNA1	customer_name	2	0.01	Low
KNA1	country_code	2	0.01	Low
KNA1	city	155	0.77	Low
LFA1	vendor_name	5	0.16	Low
LFA1	country_code	3	0.10	Low
LFA1	city	250	8.24	High
SKA1	created_by_user	4,060	2.77	Medium
SKA1	created_date	4,060	2.77	Medium
SKAT	chart_of_accounts	782,051	100.00	High
SKAT	gl_account_description	20,239	2.59	Medium
SKAT	gl_account_short_name	16,666	2.13	Medium
T001	country_code	24	3.97	Medium
T001	chart_of_accounts	16	2.64	Medium
T001	fiscal_year_variant	8	1.32	Medium

Most failures are small. The ones that matter are SKAT.chart_of_accounts, which is empty in all 782,051 rows, and LFA1.city, which is missing for 8.24 percent of vendors. The audit fields in SKA1 (created_by_user and created_date) are missing for the same number of rows, 4,060, so it is probably the same records.

Some columns with very high NULL rates were not scored because they look optional or may be legitimately blank, and need a decision from the business. They are BSEG.tax_code (100 percent NULL), BSEG.special_gl_indicator (99.998 percent), BSEG.line_item_text and cost_center_id (99.98 percent), SKA1.financial_statement_type (59.30 percent) and SKA1.balance_sheet_indicator (40.70 percent). For the last one, a blank value may simply mean that the account is not a balance sheet account. KNA1.customer_name_2 (95.32 percent) and LFA1.vendor_name2 (84.78 percent) are second name lines and are expected to be mostly empty.

4.2 Uniqueness

Eight of the ten tables contain rows that are identical in every column (Table 3). In total 72,992 rows repeat an earlier row, and 98.5 percent of them are in SKAT.

Table 3: Exact duplicate rows

Table	Duplicate rows	% of table	Rows in duplicate groups	Result	Severity
BKPF	470	0.31	921	Fail	Low
BSEG	77	0.02	154	Fail	Low
CSKS	40	1.19	79	Fail	Medium
CSKT	2	0.01	4	Fail	Low
KNA1	23	0.11	42	Fail	Low
LFA1	389	12.82	488	Fail	High
SKA1	0	0.00	0	Pass	-
SKAT	71,924	9.20	126,461	Fail	High
T001	67	11.07	92	Fail	High
T002	0	0.00	0	Pass	-

Only exact duplicates were counted. Rows that share a business key but differ in other columns were not checked, so the true level of duplication in the master tables may be higher. In T001 the duplicates shown by the profiling notebook were template company codes that each appear twice with identical values.

4.3 Validity

Table 4 shows the validity rules that could be tested from the available results.

Table 4: Validity rules

Rule	Applies to	Result	Detail
fiscal_period is between 1 and 12	BKPF	Pass	Observed values run from 1 to 12.
Amounts are greater than zero	BSEG	Pass	Minimum is 0.03; no zero or negative amounts.
Amounts stay below a plausible upper limit	BSEG	Review	Maximum is about 98.98 billion and the mean is about 250 times the median. No limit has been defined, so the large values need checking with the finance team.
Date and time values use a date or timestamp type	BKPF (entry_time), BSEG (clearing_date, payement_baseline_date), SKA1 (created_date, record_timestamp), SKAT (record_timestamp), T002 (record_timestamp)	Fail	Seven columns are stored as text. Whether the stored text is a valid date was not tested.
Flag fields use the BOOLEAN type	T002	Fail	is_deleted is stored as text.

Value-level checks, such as whether country codes, currency codes or document types come from an approved list, or whether text columns follow the expected format, were not possible because the profiling notebooks did not output value frequencies.

4.4 Consistency

Table 5 shows the consistency rules that were tested across the tables.

Table 5: Consistency rules

Rule	Result	Detail
client_id has the same type in all tables	Fail	Text in BKPF, BSEG, CSKS, CSKT, SKA1, SKAT and T001, but BIGINT in KNA1 and LFA1.
Date and time columns are stored the same way	Fail	Three different approaches are used: DATE, TIMESTAMP and text.
is_deleted has the same type in all tables	Fail	BOOLEAN in SKA1 and SKAT, text in T002.
The same field has the same name in all tables	Fail	gl_account_no (BSEG, SKA1) and gl_account_number (SKAT); regional_code (KNA1) and region_code (LFA1); customer_name_2 and vendor_name2; payement_baseline_date is misspelled.

These are low severity for the data itself, but they will cause errors or silent mismatches when the tables are joined, for example BKPF to BSEG on client and document number, or KNA1 to BSEG on client and customer.

5. Priority Issues

1. SKAT: 71,924 duplicate rows (9.20 percent) and chart_of_accounts empty in every row. This is the largest problem in the data.
2. LFA1: 389 duplicate rows (12.82 percent) and city missing for 8.24 percent of vendors.

3. T001: 67 duplicate rows (11.07 percent) among 605 company codes, plus NULLs in country_code, chart_of_accounts and fiscal_year_variant. Because this is a small reference table that other tables depend on, errors here spread further than the counts suggest.
4. SKA1 and CSKT: audit fields missing for 2.77 percent of accounts and cost center names missing for 1.05 percent of rows. CSKS has 1.19 percent duplicates.
5. BSEG: extremely large amounts that need validation, and a few NULL amounts (3 local and 92 document currency).
6. Mismatched types and names across tables, which affect every join.

6. Recommendations

As in the profiling stage, the staging tables should not be changed directly. The actions below apply to separate cleaned datasets, after each rule has been agreed.

1. Confirm the quality rules in this report with the data owners, in particular which fields are mandatory, and set an acceptable failure threshold for each.
2. Confirm the business key of each table from the SAP data dictionary and run the key uniqueness check. Then decide a de-duplication rule, starting with SKAT, LFA1 and T001, and find out whether the repeats come from repeated extracts or from change records.
3. Check at the source whether SKAT.chart_of_accounts and BSEG.tax_code are really unused or were lost during extraction. If the value can be taken from a related table, that should be agreed first and not assumed.
4. Send records that fail a mandatory-field rule to a separate review set, and do not fill the gaps with default values.
5. Check the large BSEG amounts and define an upper limit for the amount fields.
6. Convert the text date and flag columns to proper types, standardise client_id and the column names in the transformation layer, and then test that the converted dates are valid.
7. Reconcile staging row counts with the source, run the date profiling, and repeat this assessment after every load so that scores can be tracked over time.

7. Limitations

The results depend on the assumed rules in Section 2, and a different choice of mandatory fields would change the scores. All evidence comes from the profiling outputs, so format checks, approved-value checks, key uniqueness and relationships between tables (for example every BSEG document existing in BKPF) were not tested. Duplicates are exact duplicates only.

8. Conclusion

Against the rules used here, none of the ten tables passes every check. T002 and BKPF have fully populated mandatory fields, while SKAT, LFA1 and T001 have serious problems with duplicate rows and, in SKAT, an empty key attribute. Most other failures are small in share but would still affect reporting if left unchecked. The tables can be used for further processing once the duplicates are handled, the mandatory-field failures are reviewed, and the types and names are standardised.